

Practical Python Programming

Writing a Config File Checker

This is the first part of a new column on practical Python programming. In this article, the author explains how the config file of any product can be validated by using a config file checker.



This column is aimed at enthusiasts who are interested in fast development in Python, with a minimal learning period. A rudimentary level of programming expertise is assumed, apart from an ability to visualise the task to be performed and to break it down into smaller parts using standard structures of procedural programming, like *if* or *for* constructs. I will also introduce the basic knowledge required for object-based programming, based on the Python libraries to be used. The column does not assume that readers have any level of Python knowledge.

I also aim to show how one can get started with programming in Python with an easier learning, and a fast debugging cycle. Developers, testers and even managers or architects can write small pieces of useful Python code to help improve existing products and make everyone's jobs of developing and maintaining the core product easier, even if the core product is developed in some other language. Also, Python is fun to program.

Most real world products tend to have a huge set of attributes for their programs to do their jobs. These could be inputs regarding paths of installation, options for installation, parameters for program functionality and other configuration options for things like logging levels, output files, etc. While passing them through program execution parameters and through environment variables is an option, most professional programs pass and control such inputs using configuration files.

When the configuration files are correct, the programs work fine. But when they are malformed, or erroneously edited, most programs fail to function well. This may result in some aspect of the product's availability or functionality being hampered. One way to handle this would be for the programs parsing the parameters to also make a check for valid entries in the configuration files. The programs could then fail safely and instruct users about the erroneous configurations. This, however, could make the programs bulkier. Also, there may not be valid ways for server